Yugesh B

Open to new opportunities

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ABOUT

Summary

Data engineer with 4+ years across web-scraping automation, ETL/data pipelines, and GenAI-powered applications. Started in Perl-based publishing automation, moved through large-scale e-commerce scraping and Bloomberg financial data engineering at TCS, and now designs distributed data infrastructure and LLM/VLM-powered extraction pipelines at Thurro — comfortable owning a problem end-to-end, from a raw HTML page or PDF to a production system serving real analytics.

CAREER

Experience

Data Engineer
Thurro — Chennai, India
Dec 2025 – Present

- Designed and built a distributed job-execution platform (FastAPI + MySQL + APScheduler) that centrally schedules, dispatches, and tracks ~1,500 production web-scraping scripts across a multi-server AWS EC2 fleet, replacing a fragmented per-machine cron setup with zero code duplication.
- Architected a broker-free, database-polling job queue using row-level locking (SELECT ... FOR UPDATE SKIP LOCKED) with automatic crash detection and retry-with-backoff, deployed end-to-end on AWS EC2 Windows servers as auto-restarting native services (NSSM).
- Built and hardened a config-driven RAG ingestion pipeline processing 28+ financial/regulatory sources into a ClickHouse-backed vector corpus, including diagnosing a native heap-corruption crash and a cross-host n8n execution bug in production.
- Reverse-engineered a captcha-gated government vehicle-registration portal into a 5–10x faster automated data pipeline, uncovering and fixing a data-classification bug responsible for ~1.7M misclassified vehicle registrations.
- Built an LLM+VLM-powered extraction pipeline turning 40+ regulatory PDF disclosure forms across 30+ Indian insurance companies into structured financial data, using a two-pass text-then-vision verification flow.

Senior Analyst (Python Data Analyst) — Key Client: Bloomberg
Tata Consultancy Services — Chennai, India
Feb 2025 – Nov 2025

- Developed Power BI dashboards to visualize Bloomberg financial datasets, enabling data-driven decisions for stakeholders.
- Automated data cleaning (Python/pandas) and ETL pipelines for Bloomberg market data feeds, improving processing speed by 30%.
- Collaborated with Bloomberg's analytics team to optimize data-retrieval queries, reducing report generation time by 25%.
- Designed a web scraping tool (Scrapy/Selenium) to collect supplementary financial data from public sources.

Software Engineer
Lumina Datamatics — Chennai, India
May 2024 – Feb 2025

- eBay web scraping: end-to-end pipelines (Scrapy, BeautifulSoup, Selenium) against JavaScript-heavy, AJAX-based pages; cut scraping time 40% via multi-threaded crawling.
- Processed scraped data into MongoDB/MySQL; built validation/transformation scripts and integrated REST APIs for real-time sync.

Junior Software Engineer
Lumina Datamatics — Chennai, India
May 2023 – May 2024

- JXpress publishing platform: automated XML/HTML transformations using Perl and regex-based text processing.
- Python/MySQL automation for content-classification, improving workflow efficiency 25% across 500+ daily manuscripts.

Software Engineering Trainee
Lumina Datamatics — Chennai, India
May 2022 – May 2023

- Onboarded into production software engineering practices — agile delivery, code review, and on-call support.

CASE STUDIES

Featured Projects

Distributed Scraper Execution Platform

Broker-free job orchestration for a ~1,500-script scraping fleet, 221 live in production

- Built a distributed job-execution platform (FastAPI+MySQL+APScheduler) for a ~1,500-script scraping fleet — zero code duplication, scaled live usage from 20 to 221 scripts.
- Live operational CLI, two-tier CI/CD (91 tests), and an AST-based lint tool; fixed 5 real production incidents.

Python · FastAPI · SQLAlchemy 2.0 · APScheduler · MySQL · AWS EC2 · GitHub Actions

AdQvest Document Chunking & Embedding Pipeline

Config-driven RAG ingestion pipeline turning 28+ regulatory sources into a searchable vector corpus

- Built a config-driven pipeline unifying 28+ heterogeneous regulatory sources into a ClickHouse-backed RAG corpus via one YAML-configured engine.
- Redesigned serialized chunking into a concurrent, per-table-parallel architecture; diagnosed and fixed a native heap-corruption crash and a MySQL connection-leak cascade in production.

Python · MySQL · ClickHouse · AWS S3 · Docker · VoyageAI · n8n

VAHAN Vehicle Registration Data Pipeline

Reverse-engineered a captcha-gated govt. portal; found a bug behind 1.7M misclassified registrations

- Reverse-engineered an undocumented JSON API behind a captcha-gated government portal, replacing browser-automation with a 5–10x faster direct-API pipeline across 20+ report configurations.
- Root-caused and fixed a data-classification bug conflating two EV fuel types, recovering ~1.7M misclassified vehicle registrations — proved via exact month-by-month reconciliation.

Python · requests · Playwright · MySQL · pandas · Ollama · Zyte

IRDAI Insurance Disclosure Data Extraction Pipeline

Two-pass LLM+VLM pipeline extracting structured data from 40+ regulatory PDF forms, 30+ insurers

- Built an automated LLM+VLM pipeline extracting structured financial data from 40+ regulatory PDF forms across 30+ Indian insurers and 10+ fiscal quarters, eliminating manual data entry.
- Two-pass multimodal extraction (LLM text pass + VLM vision pass) with coordinate-based table reconstruction; custom compact output format (TOON) cutting token usage.

Python · PyMuPDF · Gemini API · Ollama · ClickHouse · AWS S3

Indian Law AI Chatbot

Personal project — legal Q&A assistant with semantic search over a self-refreshing 40,000-entry dataset

- Streamlit chatbot grounded in a 40,000-entry dataset covering the Supreme Court and all 24 High Courts, refreshed daily.
- Replaced a static 35K-token context dump with real semantic search — cut tokens ~30K/message; added a zero-LLM case-status router and a Groq fallback.

Python · Streamlit · Gemini API · Groq · SQLite · Plotly

Automated Document Intelligence Pipeline

Multi-engine scraping, PDF extraction & RAG pipeline for 277 sources across India and the US

- Config-driven scraping framework covering 277 site configurations from one unified codebase, with multi-engine fallback (Selenium → Playwright → HTTP).
- Document-to-vector RAG pipeline (hybrid PDF extraction, VoyageAI embeddings, Milvus) and direct SEC EDGAR ingestion for US filings; grown to ~40.5K lines of Python.

Python · Selenium · Playwright · MySQL · ClickHouse · Milvus · VoyageAI

TOOLBOX

Skills & Technologies

LANGUAGES

Python · Perl · SQL

BACKEND & APIS

FastAPI · APScheduler · REST APIs · SQLAlchemy 2.0

WEB SCRAPING & AUTOMATION

Selenium · Playwright · Scrapy · BeautifulSoup · requests

DATABASES

MySQL · MongoDB · SQLite · ClickHouse · Milvus

AI / GENAI

Gemini API · Ollama · VLMs · RAG Pipelines · VoyageAI

CLOUD & TOOLING

AWS S3/EC2 · Docker · NSSM · n8n · Zyte · GitHub Actions

CREDENTIALS

Certifications & Education

CERTIFICATIONS

- Salesforce Certified Administrator (SCA)
- Python — LinkedIn Learning
- Data Science / Machine Learning — LinkedIn Learning

EDUCATION

BCA — Computer Programming & Applications

Vel-Tech Ranga Sanku Arts College, Chennai

2018 – 2021